



Perioperative considerations in anesthesia for minimally invasive repair of pectus excavatum, Nuss procedure

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ARTICLE INFO

Keywords:

Pectus excavatum
Nuss procedure
Perioperative
Analgesia
ERAS

ABSTRACT

Pectus excavatum (PE) is a common anterior chest wall deformity that affects the heart and lungs depending on the severity of compression. The Nuss procedure, a minimally invasive repair of pectus excavatum, has evolved over the years with thoroscopic assistance and minimal incision. Despite improved surgical techniques, pain and nausea are often the most common factors determining hospital length of stay. This review will explore the preoperative, intraoperative, and postoperative anesthetic considerations necessary for improving patient outcomes, reducing surgical stress, and shortening hospital stays for patients undergoing the Nuss procedure.

Background

Pectus excavatum (PE) is the most common chest wall deformity in children and young adults, progressing rapidly with the onset of puberty. It occurs in up to 1 in 400 male and 1 in 2000 female births.^{1,2} Characterized by concavity of the anterior chest wall—meaning the chest appears sunken in—PE causes numerous symptoms along a spectrum of severity including chest pain, fatigue, dyspnea, decreased exercise tolerance, and issues relating to body image and self-concept.³ The minimally invasive repair of pectus excavatum (MIRPE)—or Nuss procedure—was introduced in 1998 and has largely replaced the open Ravitch procedure as the standard in pectus excavatum repair.⁴ Due to its widespread adoption and high technical success in pediatric surgery, special anesthetic considerations are necessary when caring for pediatric patients undergoing the Nuss procedure.

Preoperative considerations

Risk stratification

Pediatric patients undergoing the Nuss repair procedure must receive preoperative risk stratification for effective and safe anesthetic and surgical planning. The Haller index (HI) is typically used to measure the severity of pectus excavatum (PE). The HI is calculated by dividing

the widest transverse diameter of the thoracic cavity at the deepest point of the defect by the anterior-posterior diameter at the same point using CT imaging.⁵ A HI greater than 3.25 indicates severe deformity and has historically justified surgical repair. However, this measurement fails to fully account for the range of functional limitations associated with PE. Recent efforts have been made to develop more descriptive and accurate indices and reduce radiation exposure, but the CT-derived HI remains the standard for quantifying PE severity.^{5,6}

Patients with PE often exhibit reduced total lung capacity (TLC) and forced vital capacity (FVC), but increased forced expiratory volume in 1 second (FEV1) on pulmonary function tests (PFT). However, most patients still have normal PFT results before and after surgery.^{1,2,7} Obstructive patterns have also been observed and are related to patient-specific factors.⁷ When present, the severity of restrictive lung disease may correlate with the HI.⁸

The cardiovascular effects of PE are related to compression of the right ventricle (RV) and subsequent impairment of left ventricular (LV) function due to ventricular interaction. Functional impairment is often not apparent until adolescence. Cardiopulmonary exercise testing may be a feasible and reliable method for measuring decreased exercise tolerance in pediatric patients with PE, but it is not currently practical or routinely used in practice.² Transthoracic echocardiography (TTE) is routinely performed during the perioperative evaluation of pediatric patients with PE, although this practice varies among institutions.⁷ TTE

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<https://doi.org/10.1016/j.sempedsurg.2024.151459>

is widely used, minimally invasive, and highly sensitive in detecting the most common echocardiographic abnormalities in pediatric patients with PE, such as decreased end diastolic RV volume, increased RV ejection fraction, mitral valve prolapse, and regurgitation.^{2,7} Abnormalities in cardiac output and index have also been observed in this population, although the overall incidence of abnormal TTE findings is low.² Some evidence supports the use of thoracic CT or MRI alone to accurately assess cardiac architecture, but TTE is still preferred in many institutional preoperative protocols.¹

Premedication

When premedicating pediatric patients for the Nuss procedure, it is important to consider post-operative nausea and vomiting (PONV), pain and anxiety. Surgical repair of PE, regardless of the approach used, is associated with severe pain. Therefore, a multimodal approach to analgesia should be taken before, during, and after the surgery. It is increasingly emphasized the importance of developing institution-wide pain management protocols for the pediatric Nuss procedure that require buy-in from patient families, as well as anesthesia, surgical, nursing, and allied health staff.^{9,10}

The pre-emptive initiation of gabapentin at home, along with pre-operative acetaminophen, long-acting oral opioids, and non-steroidal anti-inflammatory drugs (NSAIDs) on the day of surgery, have been well-described and appear to be common practice.^{11,12} The use of gabapentin both before and after the surgery has been shown to reduce the total amount of opioids administered perioperatively. Additionally, premedication with benzodiazepines has shown potential benefits in reducing postoperative subjective pain scores, highlighting the relationship between anxiety and pain in pediatric patients and underscoring the importance of providing anxiolysis.^{13,14}

PONV seems to affect pediatric patients undergoing the Nuss procedure more than other patient populations, with some observational studies reporting that up to 60 % of adolescent Nuss procedure patients experience nausea with or without vomiting after surgery. This may be related to high opioid requirements in the postoperative period. Therefore, early and aggressive PONV prophylaxis using ondansetron and dexamethasone is warranted.¹⁵ While there is limited support for using transdermal scopolamine for PONV prophylaxis in this patient population, some studies suggest that it is safe and effective in pediatric patients when applied before surgery. However, caution should be exercised when using transdermal scopolamine due to the high reported incidence of anticholinergic side effects in pediatric patients.¹⁶

Intraoperative considerations

Airway management

Achieving effective lung isolation is crucial for ensuring the safety and success of the Nuss procedure. One-lung ventilation (OLV) in children presents specific challenges, including endotracheal tube (ETT) sizing, design, availability, and compatibility with adjunct devices. Since most patients undergoing the Nuss procedure are at least 8 years old, their airway anatomy and physiology are similar to that of adults.¹⁷ However, the reduced size of the single-lumen ETT and left endobronchial intubation limit the flexibility in alternating lung isolation. In such cases, intra- or extraluminal placement of a bronchial blocker remains a feasible option, particularly for patients who are too small or have airway abnormalities that prevent the safe placement of a double-lumen ETT (DLT).¹⁸ Left-sided DLTs are available in sizes as small as 26 Fr and enable safe and effective OLV for the majority of patients undergoing the Nuss procedure. Some case series have investigated the utility of supraglottic airways in the anesthetic management of the Nuss procedure, and they have concluded that these airways may provide a safe and feasible alternative to endotracheal intubation with modifications to surgical technique. However, endotracheal intubation remains the

standard of care.^{19,20}

Multimodal analgesia

Despite its minimally invasive nature, postoperative pain following the Nuss procedure remains challenging. While there is no universally accepted pain management strategy, there is growing support for multimodal analgesia. Regional anesthesia, performed under sedation or general anesthesia, is common. Numerous cases and studies describe success with thoracic epidural catheters, paravertebral and erector spinae plane (ESP) single-shot blocks or catheters, PECS II and serratus anterior blocks, wound catheters, and intercostal nerve blocks or ablations.^{9,21-23}

Historically, thoracic epidural analgesia (TEA) has been effective for managing postoperative pain in pediatric patients undergoing the Nuss procedure. A recent multicenter observational study suggests that it remains the most utilized regional technique.²⁴ However, TEA has limitations including variable efficacy, rare but severe complications, high rates of postoperative nausea and vomiting (PONV), resource-intensive postoperative management, longer hospital stays on average, and delayed ambulation compared to patients who do not receive neuraxial analgesia.^{10,12,24-27}

The paravertebral block has been identified as a safe and beneficial alternative to TEA in this patient population, but its efficacy is limited by advanced technical requirements and the need for bilateral placement.²⁸ Similarly, bilateral ESP blocks and catheters have been explored as safer and simpler alternatives to TEA. Supporting evidence indicates fewer complications, lower opioid requirements (morphine milligram equivalents, or MME), and earlier hospital discharge.^{29,30}

There are limited strong data suggesting the superiority of one regional technique over another in this patient population. Treatment decisions should be made after considering patient and institution-specific factors. Cryoanalgesia is an emerging adjunct in the Nuss procedure that has demonstrated high efficacy and a favorable safety profile.^{26,31} Despite lengthening the operative time, surgical cryoablation of the intercostal nerves is associated with improved subjective pain scores, decreased incidence of PONV and constipation, and an earlier transition from parenteral medications.^{9,32,33} Compared to patients receiving TEA, those who receive cryoanalgesia require fewer MMEs, are discharged home sooner, and incur lower total healthcare costs.^{26,34,35} One notable limitation of this technique is the time to onset of effect: Wallerian degeneration of intercostal nerves may take up to 24 hours, meaning cryoablation alone may be insufficient for analgesia in the immediate postoperative period.³⁶ The incorporation of cryoablation into a multimodal pain management strategy may offer a solution. In addition to cryoablation alone, ESP, paravertebral, and intercostal nerve blocks have been described as decreasing hospital stays and improving patient-centric outcomes.^{14,36}

Numerous studies have explored the benefits of parenteral opioids, steroidal and nonsteroidal anti-inflammatory drugs, and NMDA-receptor antagonistic drugs such as ketamine in pain management for pediatric patients undergoing the Nuss procedure. There is evidence supporting the intraoperative administration of ketamine to reduce postoperative opioid requirements and subsequently reduce opioid-related side effects.¹⁵ Similarly, intraoperative methadone may decrease or eliminate postoperative opioid requirements but does not appear to improve subjective pain scores, hospital length of stay, time to ambulation, or time to diet advancement.^{11,37} The combination of these two drugs may produce a synergistic reduction in postoperative opioid consumption and reported pain scales.³⁸

Dexmedetomidine lowers postoperative pain scores and decreases emergence delirium in pediatric patients undergoing thoracoscopic surgery and may be of utility for the Nuss procedure.³⁹ Dexamethasone administered intraoperatively at low doses has been shown to reduce postoperative pain scores and the incidence of PONV.³⁸ If patients have not received COX-2 inhibitors preoperatively, intraoperative use of

ketorolac is an effective and safe analgesic adjunct.⁹ Incorporating and adhering to a comprehensive, multimodal analgesia plan contributes to superior patient outcomes and enhanced recovery after surgery (ERAS).^{10,40} Table 1 highlights the various regional anesthesia modalities for Nuss procedures.

Access, monitoring, and other management

While much of the literature on anesthetic considerations for Nuss procedures focuses on pain and ventilatory management, other essential intraoperative factors must also be considered. The placement of pectus bars is typically atraumatic, but there is a risk of injury to intrathoracic structures that could result in serious, life-threatening hemorrhage.⁴¹ Morbidity and mortality are rare with the Nuss procedure, and while transfusion of blood products is uncommon, additional preparation and risk stratification must be made for patients with a history of prior chest surgery, pericarditis, or severe pectus excavatum deformity. In high-risk patients or those with a history of cardiac anomalies, transesophageal echocardiography may be beneficial for real-time monitoring of volume status and cardiac performance.^{41,42} Invasive blood pressure monitoring and two peripheral intravenous access points are sufficient for most cases.¹⁷ Maintenance of anesthesia is typically via inhalational agents. Thermoregulation is complicated by radiant heat loss, pneumothorax gas flows, and cryoanalgesia; although there is conflicting evidence on the impact of cryoanalgesia on inadvertent perioperative hypothermia.^{43,44} Efforts to minimize heat loss—including low fresh gas flows, application of a heat and moisture exchanger to the ventilator circuit, and active warming—should be employed as precautionary measures.

Following bar placement and early closure, iatrogenic pneumothorax is addressed by performing a Valsalva maneuver. Avoidance of coughing or straining is preferred due to the risk of subcutaneous emphysema and residual pneumothorax. Deep extubation is ideal and preferred in the absence of contraindications.¹⁷

Postoperative considerations

For most patients, discharge from the post-anesthesia care unit to an inpatient surgical floor appears safe, and intensive care unit admission should be reserved for patients with complicated intraoperative courses or decompensated comorbidity.⁹ Pain management and opioid-related side effects are the two most common factors for prolonged hospitalization following the Nuss procedure, emphasizing the importance of multimodal analgesia. Common modalities include opioid-based intravenous patient-controlled analgesia (IVPCA), thoracic epidural analgesia (TEA), and infusion of local anesthetic medications into erector spinae plane (ESP), paravertebral, or wound catheters placed intraoperatively.⁴⁵

Opioid therapy remains a mainstay in the management of postoperative pain following the Nuss procedure. Patients commonly receive IVPCA in recovery and on postoperative day 0 (POD 0). This can be transitioned directly to enteral opioids as early as POD 1, and a combination of both immediate and sustained-release formulations has been described as safe and effective in this patient population in the inpatient setting.³⁸ Opioid therapy is associated with potential hospital stay-prolonging side effects, and the use of opioids should be balanced with other patient-specific factors.

Addition of adjuncts such as dexamethasone and clonidine to local agents in ESP and paravertebral infusions effectively reduces the total MME requirement.⁴⁶ Clonidine has been administered either intravenously or transdermally postoperatively with good effect, reducing pain and providing anxiolysis. Furthermore, transdermal clonidine is associated with a lower incidence of hemodynamic disturbances compared to bolus dose intravenous clonidine or continuous dexmedetomidine infusion.⁴⁷ Other non-narcotic adjuvants, including NSAIDs, anticonvulsants, and sedatives, have been shown to reduce MME requirement postoperatively. Transdermal lidocaine has been demonstrated to

Table 1
Regional anesthesia techniques for patients undergoing minimally invasive pectus excavatum repair.

Technique	Advantages	Disadvantages
Thoracic Epidural Analgesia (TEA)	Most effective pain management Early return of bowel function	Longer hospital stays Hemodynamic instability PONV
Paravertebral Block	Effective pain management Safer than TEA	Advanced technique Bilateral placement necessary
Erector Spinae Plane Block	Safe and effective Few complications compared to other regional techniques	Insufficient strong data supporting use in this population
Cryoanalgesia	Safe and effective Long duration of action	Delayed onset of effect

reduce subjective pain scores.^{11,25,45} Diazepam can be utilized for its anxiolytic, antiemetic, and muscle-relaxant properties and can be continued orally as an inpatient.³⁸ There is limited evidence suggesting that behavioral techniques such as hypnosis and integrative medical practices may effectively manage pain following the Nuss procedure.⁴⁸ Some institutions have shown that initiation of a multimodal ERAS pain protocol may accelerate hospital discharge as early as postoperative day one (14).

As mentioned previously, postoperative nausea and vomiting (PONV) remains a significant concern following the Nuss procedure in pediatric patients. Early mobilization and the utilization of 5HT-3 receptor antagonists such as ondansetron or granisetron, as well as antihistamines such as diphenhydramine or promethazine, are effective treatment options for patients experiencing severe nausea or vomiting.³⁸ In the absence of detrimental side effects, a transdermal scopolamine patch applied preoperatively can be left in place to reduce nausea and/or vomiting for up to 72 hours.

Conclusion

From an anesthetic perspective, the goal of the Nuss procedure is to improve patient outcomes, reduce surgical stress, and shorten hospital stays. ERAS protocols, starting preoperatively, require a team approach, including patient and family education and institutional buy-in. Multimodal strategies with oral medications, intraoperative opioids, antiemetics, and opioid-sparing drugs like ketamine, dexmedetomidine, lidocaine, and NSAIDs aim to reduce opioid use and accelerate the transition to oral medication.

References

1. Obermeyer RJ, Goretsky MJ. Chest wall deformities in pediatric surgery. *Surg Clin North Am.* 2012;92:669–684.
2. Obermeyer RJ, Cohen NS, Jaroszewski DE. The physiologic impact of pectus excavatum repair. *Semin Pediatr Surg.* 2018;27:127–132.
3. Kirupaharan S, Briatico D, Robinson T, Fitzgerald P, Walton JM. Postoperative management of pediatric patients undergoing minimally invasive repair of pectus excavatum: Where are we now? *J Pediatr Surg.* 2022;57:927–931.
4. Du X, Mao S, Cui J, et al. Use of laryngeal mask airway for non-endotracheal intubated anesthesia for patients with pectus excavatum undergoing thoracoscopic Nuss procedure. *J Thorac Dis.* 2016;8:2061–2067.
5. Gosztyla CE, Petrosyan M, Kane T, et al. Mini thoracic CT adequately determines Haller index and decreases radiation exposure in children with pectus excavatum. *J Pediatr Surg.* 2022;57:1076–1078.
6. Sujka JA, Peter ST SD. Quantification of pectus excavatum: Anatomic indices. *Semin Pediatr Surg.* 2018;27:122–126.
7. Ganesu O, LaRusso K, St-Louis E, et al. The utility of echocardiography and pulmonary function testing in the preoperative evaluation of pectus excavatum. *J Pediatr Surg.* 2022;57:1561–1566.
8. Jeong JY, Ahn JH, Kim SY, et al. Pulmonary function before and after the Nuss procedure in adolescents with pectus excavatum: correlation with morphological subtypes. *J Cardiothorac Surg.* 2015;10:37.
9. Cockrell HC, Hrachovec J, Schnuck J, Nchinda N, Meehan J. Implementation of a Cryoablation-based Pain Management Protocol for Pectus Excavatum. *J Pediatr Surg.* 2023;58:1239–1245.

10. Singhal NR, Jones J, Semenova J, et al. Multimodal anesthesia with the addition of methadone is superior to epidural analgesia: a retrospective comparison of intraoperative anesthetic techniques and pain management for 124 pediatric patients undergoing the Nuss procedure. *J Pediatr Surg.* 2016;51:612–616.
11. Grisales PA, Rauh JL, Benfield AL, et al. Raising the bar: multimodal analgesia with transdermal lidocaine for nuss repair of pectus excavatum decreases length of stay and opioid use. *J Pediatr Surg.* 2023;58:2244–2248.
12. Ganesu O, Emil S, Saint-Martin C, et al. Postoperative pain following minimally invasive repair of pectus excavatum: a descriptive study. *J Pediatr Surg.* 2022;57:918–926.
13. Fenikowski D, Tomaszek L, Mazurek H, Gawron D, Maciejewski P. The effects of gabapentin on post-operative pain and anxiety, morphine consumption and patient satisfaction in paediatric patients following the ravitch procedure-a randomised, double-blind, placebo-controlled, phase 4 trial. *J Clin Med.* 2022;11.
14. DiFiore JW, Robertson JO, Chhabada S, et al. Next day discharge after the Nuss procedure using intercostal nerve cryoablation, intercostal nerve blocks, and a perioperative ERAS pain protocol. *J Pediatr Surg.* 2022;57:213–218.
15. Frawley G, Frawley J, Cramer J. A review of anesthetic techniques and outcomes following minimally invasive repair of pectus excavatum (Nuss procedure). *Paediatr Anaesth.* 2016;26:1082–1090.
16. Pergolizzi JV, Raffa R, Taylor R. Prophylaxis of postoperative nausea and vomiting in adolescent patients: a review with emphasis on combination of fixed-dose ondansetron and transdermal scopolamine. *J Drug Deliv.* 2011;2011, 426813.
17. Mavi J, Moore DL. Anesthesia and analgesia for pectus excavatum surgery. *Anesthesiol Clin.* 2014;32:175–184.
18. Templeton TW, Piccioni F, Chatterjee D. An update on one-lung ventilation in children. *Anesth Analg.* 2021;132:1389–1399.
19. Liu R-S, Cui F-X, Li Y-J, et al. Retrospectively analyze and compare the efficacy and safety of thoracoscopic-assisted Nuss repair of pectus excavatum under intubation anesthesia and non-intubation anesthesia. *J Thorac Dis.* 2022;14:4031–4043.
20. Yu P, Wang G, Zhang C, et al. Clinical application of enhanced recovery after surgery (ERAS) in pectus excavatum patients following Nuss procedure. *J Thorac Dis.* 2020;12:3035–3042.
21. Beltran R, Veneziano G, Bhalla T, et al. Postoperative pain management in patients undergoing thoracoscopic repair of pectus excavatum: A retrospective analysis of opioid consumption and adverse effects in adolescents. *Saudi J Anaesthesia.* 2017;11:427–431.
22. Sertcakacilar G, Kose S. Bilateral PECS II block is associated with decreased opioid consumption and reduced pain scores for up to 24 hours after minimally invasive repair of pectus excavatum (Nuss Procedure): a retrospective analysis. *J Cardiothorac Vasc Anesth.* 2022;36:3833–3840.
23. Tore Altun G, Arslantas MK, Corman Dincer P, Aykac ZZ. Ultrasound-guided serratus anterior plane block for pain management following minimally invasive repair of pectus excavatum. *J Cardiothorac Vasc Anesth.* 2019;33:2487–2491.
24. Gurria JP, Simpson B, Tuncel-Kara S, et al. Standardization of clinical care pathway leads to sustained decreased length of stay following Nuss pectus repair: A multidisciplinary quality improvement initiative. *J Pediatr Surg.* 2020;55:2690–2698.
25. Yang W, Ming Y-C, Kau Y-C, et al. A comparison of parecoxib and thoracic epidural analgesia for postoperative analgesia following Nuss procedure. *J Pediatr Surg.* 2015;50:2032–2034.
26. Daemen JHT, de Loos ER, Vissers YLJ, Bakens MJAM, Maessen JG, Hulsewe KWE. Intercostal nerve cryoablation versus thoracic epidural for postoperative analgesia following pectus excavatum repair: a systematic review and meta-analysis. *Interact Cardiovasc thorac Sur.* 2020;31:486–498.
27. Nakamura K, Sugita A, Sekiya S, et al. Paths of thoracic epidural catheters in children undergoing the Nuss procedure for pectus excavatum repair. *J Anesth.* 2022;36:335–340.
28. Loftus PD, Elder CT, Russell KW, et al. Paravertebral regional blocks decrease length of stay following surgery for pectus excavatum in children. *J Pediatr Surg.* 2016;51:149–153.
29. Wu S, Wu J, Zhang X. Application of an ultrasound-guided bilateral erector spinae plane block after the Nuss procedure for pectus excavatum in children: a retrospective cohort study with propensity score matching. *Fron Pediatr.* 2023;11, 1201604.
30. Walter CM, Lee CS, Moore DL, et al. Retrospective study comparing outcomes of multimodal epidural and erector spinae catheter pain protocols after pectus surgery. *J Pediatr Surg.* 2023;58:397–404.
31. Lai K, Eldredge RS, Zobel M, Hargis-Villanueva A, Ostlie A, Padilla BE. Intercostal nerve cryoablation for postoperative pain control in pediatric thoracic surgery: a scoping review. *J Laparoendosc Adv Sur Techn Part A.* 2023;33:994–1004.
32. Aiken TJ, Stahl CC, Lemaster D, et al. Intercostal nerve cryoablation is associated with lower hospital cost during minimally invasive Nuss procedure for pectus excavatum. *J Pediatr Surg.* 2021;56:1841–1845.
33. Perez MN, Ferrantella A, Johnson KN, Goldstein SD. Intercostal spinal nerve cryoablation for analgesia following pectus excavatum repair. *Semin Pediatr Surg.* 2024;33, 151382.
34. Keller BA, Kabagambe SK, Becker JC, et al. Intercostal nerve cryoablation versus thoracic epidural catheters for postoperative analgesia following pectus excavatum repair: Preliminary outcomes in twenty-six cryoablation patients. *J Pediatr Surg.* 2016;51:2033–2038.
35. Zeineddin S, Goldstein SD, Linton S, et al. Effectiveness of one minute per level intercostal nerve cryoablation for postoperative analgesia after surgical correction of pectus excavatum. *J Pediatr Surg.* 2023;58:34–40.
36. Aranda-Valderrama P, Greenberg RS, Vecchione TM, Finney AG, Prichett LM, Vanderhoek SM. Combined erector spinae plane block with surgical intercostal nerve cryoablation for Nuss procedure is associated with decreased opioid use and length of stay. *Reg Anesth Pain Med.* 2024;49:248–253.
37. Froehling NM, Martin JA, Miles MVP, et al. Intraoperative methadone reduces postoperative opioid requirements in Nuss procedure for pectus excavatum. *Am Surg.* 2023;89:984–989.
38. Singhal NR, Jerman JD. A review of anesthetic considerations and postoperative pain control after the Nuss procedure. *Semin Pediatr Surg.* 2018;27:156–160.
39. Zhou ZG, Liu R, Tan HL, Ji XY, Yi XL, Song JF. The application of dexmedetomidine combined with dezocine in thoracoscopic radical resection of lung cancer and its effect on awakening quality of patients. *Eur Rev Med Pharmacol Sci.* 2019;23:7694–7702.
40. Zacha S, Andrzejewska A, Jastrzebska-Ligocka B, et al. Intercostal nerve cryoanalgesia in the treatment of pain in patients operated on by the modified Nuss method with the BackOnFeet application-a new strategy to improve outcomes. *Front Pediatr.* 2022;10, 1069805.
41. Kovács T, Pásztor G, Rieth A. Internal mammary artery compression after pectus excavatum repair does not increase risk of hemorrhagic complications in pediatric patients. *Front Pediatr.* 2020;8, 619065.
42. Hebra A, Kelly RE, Ferro MM, Yüksel M, Campos JRM, Nuss D. Life-threatening complications and mortality of minimally invasive pectus surgery. *J Pediatr Surg.* 2018;53:728–732.
43. Bundrant NT, Sayrs LW, Ostlie D, et al. Infectious complications of intercostal nerve cryoablation mediated by perioperative hypothermia during pediatric Nuss procedure. *J Pediatr Surg.* 2022;57:1083–1086.
44. Carter M, Inge M, Zeineddin S, et al. Measurement and Thermodynamic modeling of energy flux during intercostal nerve cryoablation. *J Surg Res.* 2024;293:231–238.
45. Muhly WT, Maxwell LG, Cravero JP. Pain management following the Nuss procedure: a survey of practice and review. *Acta Anaesthesiol Scand.* 2014;58:1134–1139.
46. Visoiu M, Sadhasivam S. A retrospective study comparing outcomes of paravertebral clonidine infusion for pediatric patients undergoing nuss procedure. *Children (Basel, Switzerland).* 2023;10.
47. Neil MJ. Clonidine: clinical pharmacology and therapeutic use in pain management. *Curr Clin Pharmacol.* 2011;6:280–287.
48. Manworren RCB, Anderson MN, Girard ED, et al. Postoperative pain outcomes after nuss procedures: comparison of epidural analgesia, continuous infusion of local anesthetic, and preoperative self-hypnosis training. *J Laparoendosc Adv Sur Techn Part A.* 2018;28:1234–1242.